

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Fullstack Engineering at Bridgeline”

Submitted in partial fulfillment for the award of degree (21INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Shreyas M J Kalkur (1DS21AI048)

Conducted at

BRIDGELINE[®]
digital

Bridgeline Digital, Inc

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**Fullstack Engineering at Bridgeline**” carried out by **Shreyas M J Kalkur [1DS21AI048]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

Signature of Reporting Head

Mr. Jayaram M V

Director,
Bridgeline

Signature of Internship Guide

Prof. Anupama V P

Dept. of AIML
DSCE

Signature of HoD

Dr. Vindhya P Malagi

Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Shreyas M J Kalkur**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in. This report is submitted in partial fulfillment of the requirements for the award of bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date: 11/04/2025

Place: Bangalore

USN: 1DS21AI048

NAME: Shreyas Kalkur

OFFER LETTER



Internship Offer

Shreyas M J Kalkur

On behalf of **Bridgeline Digital**, I am pleased to inform that you have been selected as an Intern. The position is for a **Software developer** intern.

This position is scheduled to begin from 1st October 2024 and will be a 6 months internship opportunity ending on 31st March 2025.

During your temporary employment with **Bridgeline Digital**, you may have access to trade secrets and confidential or proprietary business information belonging to **Bridgeline Digital**. By accepting this offer, you acknowledge that this information must remain confidential and agree to refrain from using it for your own purposes or disclosing it to anyone outside of **Bridgeline Digital**. Also, you agree that upon completion of your internship, you will promptly return any company-issued property and equipment along with information and documents belonging to the company.

Sincerely,

Jayaram M V
Director

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important people. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Sriram Kumar Govindaraj**, for guiding me during the period of internship.

We express our deep and profound gratitude to our Internship Guide, **Prof. Anupama V P**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would not have been possible.

Shreyas M J Kalkur [1DS21AI048]

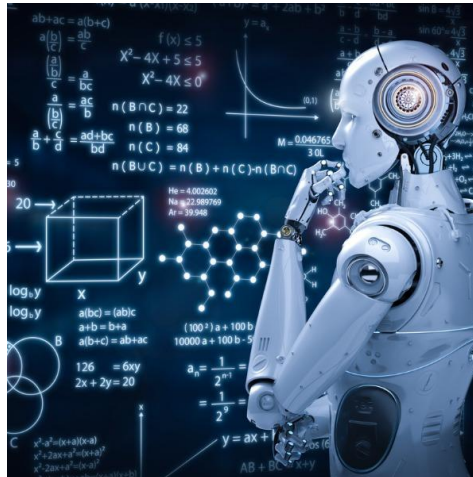
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CHAPTER 1

INTRODUCTION

1.1 Introduction to Machine Learning



Machine Learning (ML) is a branch of artificial intelligence focused on enabling computers to learn patterns and make predictions from data without being explicitly programmed for every task. By leveraging statistical techniques and algorithms, ML systems can automatically improve their performance as they are exposed to more data. From spam filtering and image recognition to recommendation engines and autonomous vehicles, ML underpins many of today's intelligent applications and services.

ML workflows typically involve collecting and preprocessing large datasets, selecting relevant features, training models on historical data, and evaluating their accuracy. Common algorithmic approaches include supervised learning (training on labeled data), unsupervised learning (discovering hidden structures in unlabeled data), and reinforcement learning (optimizing actions based on rewards). Effective ML also requires careful attention to data quality, feature engineering, hyperparameter tuning, and continuous monitoring in production to ensure models remain robust and unbiased.

Why Machine Learning is Important

- o **Automation & Efficiency:** ML automates complex tasks—such as anomaly detection or document classification reducing manual effort and speeding up processes.
- o **Data-Driven Decision-Making:** By uncovering hidden patterns and trends, ML empowers organizations to make informed strategic decisions.
- o **Personalization:** Tailors products and services (e.g., recommendations, targeted marketing) to individual user preferences, improving engagement and satisfaction.

- o **Predictive Analytics:** Forecasts future outcomes—like demand forecasting or risk assessment—allowing proactive measures and resource optimization.
- o **Innovation & Competitive Advantage:** Drives new product features, smarter services, and operational improvements that differentiate organizations in crowded markets.

ML is not merely about choosing an algorithm; it encompasses the entire lifecycle: data ingestion and cleaning, feature extraction, model selection and training, validation and testing, deployment in production environments, and ongoing performance monitoring (often supported by MLOps practices). Organizations often partner with ML service providers or build in-house teams to handle specialized tasks such as data engineering, model interpretability, scalability, and compliance with data-privacy regulations. These comprehensive efforts ensure ML solutions deliver reliable, actionable insights while adapting to evolving data and business requirements.

1.2 HawkSearch (AI-Powered Site Search)



HawkSearch is Bridgeline’s AI-driven site search and merchandising platform designed to deliver highly relevant, personalized search experiences across e-commerce and content sites. It combines machine-learning-based relevance tuning with real-time indexing, unified search modalities, and robust merchandising controls to boost traffic, conversions, and average order values

- o **Real-Time Indexing & Traffic Builder:** Continuously crawls and indexes site content and auto-generates dynamic landing pages to capture high-quality traffic.

- o **AI-Based Relevance Tuning:** Learns from user interactions to refine autocomplete suggestions, synonyms, and ranking models for more accurate results.

o Unified Search (Concept, Image & Keyword): Integrates Concept and Image Search alongside traditional keyword queries, enabling content-driven discovery and reducing zero-result sessions.

o Faceted Navigation & Merchandising Tools: Supports dynamic filters (e.g., by category, price) and allows marketers to manually promote or demote results, previewing campaign rules in real time.

o Multi-Site Management (MSM): Centralizes search and merchandising administration across multiple websites while preserving local customizations for brands or regions.

o Analytics & Performance Alerts: Provides dashboards with top search terms, zero-result counts, conversion metrics, and automated notifications for trending or underperforming keywords.

o Together, HawkSearch enables organizations to:

1. Capture and convert more traffic by surfacing hyper-relevant content.
2. Boost average order values through AI-powered recommendations and product comparisons.
3. Streamline search management across global and local sites with centralized controls.
4. Gain actionable insights to continuously refine search strategies and merchandising campaigns.

During my internship, I integrated HawkSearch's REST APIs into the React-based frontend, built custom query pipelines to support concept and image search, and developed analytics widgets in the eCommerce 360 dashboard to visualize search performance and zero-result trends.

1.3 GRC (Governance, Risk, and Compliance)



Figure 1.1: HawkSearch B2B Search App for Salesforce

Bridgeline’s AI-driven HawkSearch “B2B Search App” for Salesforce, illustrating how intelligent site search drives revenue growth by balancing three core metrics—traffic, conversion, and average order value (AOV). At the center of the diagram, Revenue is propelled by:

1. **Traffic:** Attracting qualified visitors through real-time indexing and dynamically generated landing pages.
2. **Conversion:** Turning browsers into buyers via AI-tuned relevance, autocomplete suggestions, and personalized search results.
3. **Average Order Value (AOV):** Increasing basket size with targeted merchandising, product comparisons, and cross-sell recommendations.
4. **Seamless Salesforce Integration** using the CloudCraze connector for unified B2B commerce workflows
5. **12+ Years of B2B Expertise**, reflecting a proven track record of customer success
6. **Marketer-Friendly UI** with drag-and-drop merchandising controls and preview capabilities
7. **Codeless Workflow Automation**, enabling non-technical users to define promotion rules
8. **Competitive, Usage-Based Pricing** aligned to site performance and traffic volume
9. **Customer & Pricing Entitlements**, supporting account-specific catalogs and negotiated rates

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Bridgeline Digital, Inc

BRIDGELINE

2.1 Introduction

Bridgeline Digital, Inc. is a Nasdaq-listed MarTech and Digital Experience Platform (DXP) leader founded in 2000 and headquartered in Woburn, Massachusetts. The company empowers enterprises to maximize online revenue by growing site traffic, improving conversion rates, and boosting average order value through a unified suite of solutions—ranging from content management and eCommerce to AI-powered site search and advanced analytics. Under the leadership of President & CEO Dr. Ari Kahn, Bridgeline serves over 2,500 clients across more than 100 countries in industries such as retail, manufacturing, financial services, and healthcare

Services Offered:

- **Digital Experience Platform (Unbound):** Seamlessly integrates CMS, eCommerce, marketing automation, SEO workflows, and personalization into a single platform.
- **AI-Powered Site Search & Merchandising (HawkSearch):** Delivers real-time indexing, machine-learning relevance tuning, faceted navigation, and codeless merchandising controls.
- **Analytics & Business Intelligence (DataBravo/Orchestra):** Provides interactive dashboards, automated alerts, and predictive insights on traffic, conversions, and average order value.
- **Digital Engagement & Professional Services:** Offers strategic consulting, UX/design, implementation, managed services, and ongoing support to optimize digital initiatives.
- **Flexible Deployment & Infrastructure:** Supports cloud-based SaaS, managed hosting, or on-premise installations with scalable, secure architecture.

Key Value Propositions:

- **Holistic Digital Commerce Suite:** Unifies content management, eCommerce, site search, merchandising, and analytics into a single, cohesive platform—eliminating tool sprawl and simplifying workflows.
- **Proven Performance:** Customers routinely see double-digit uplifts in key metrics—site traffic, conversion rates, and average order values—through targeted optimization and data-driven strategies.
- **AI-Powered Personalization & Search:** HawkSearch leverages machine-learning relevance tuning and no-code merchandising controls to deliver highly relevant, personalized experiences that boost engagement and sales.
- **Actionable Insights & Dashboards:** The eCommerce 360 dashboard centralizes KPIs—traffic trends, conversion performance, and order value—and surfaces optimization opportunities via real-time reporting and alerts.
- **Flexible Deployment Options:** Available as cloud-SaaS, managed hosting, or on-premise installs, with automated upgrades and minimal IT overhead for rapid time-to-value.

Innovative 5D Working Process:

Bridgeline’s proven five-phase approach ensures rapid ROI and sustained success:

Discover: Conduct stakeholder interviews, requirements gathering, and digital-maturity assessments.

Design: Architect the solution—Unbound modules, HawkSearch pipelines, and system integrations.

Develop: Configure the platform, build custom features, and perform thorough testing.

Deploy: Manage data migration, user training, go-live support, and change management.

Drive: Monitor performance via analytics, optimize search relevancy and campaigns, and deliver ongoing enhancements.

Career Opportunities and Culture:

Bridgeline Digital fosters a collaborative, growth-oriented environment that attracts innovative thinkers and passionate doers. Team members benefit from continuous learning programs, transparent communication, and a customer-first ethos—empowering individuals to drive meaningful impact and grow their careers within a supportive culture.

Mission: Grow online revenue by increasing site traffic, converting visitors into buyers,

and raising average order values.

Vision: To be the world’s premier digital-experience and MarTech partner—empowering enterprises through AI-driven innovation and data-backed insights.

Awards and Recognitions:

- Named “Moblico Partner of the Year” for innovation in mobile search technology.
- Recognized as a Top Performer in the Customer Success Report for Enterprise Search Software.
- Included in Gartner’s Digital Commerce Vendor Guide for multiple consecutive years.
- Recipient of 200+ industry awards, including Web Marketing Association Web Awards, Interactive Media Awards, W³ Awards, and Horizon Interactive Awards.

2.2 Founders

Thomas .L. Massie– Founder & Former President & CEO



Thomas L. Massie founded Bridgeline Digital in August 2000 and led its evolution from a custom services firm into a market-leading digital engagement SaaS platform. As President and CEO through 2015, he launched the iAPPS solution—deeply integrating web content management, eCommerce, email marketing, analytics, and social media management—driving rapid recurring revenue growth, a successful IPO, and more than 200 industry awards. A veteran entrepreneur, Massie previously founded Mass Microsystems and Focus Enhancements, taking both companies to Inc. 500 rankings and public offerings. Renowned for his strategic vision and hands-on leadership, he expanded Bridgeline’s global footprint, forged partnerships with Fortune 1000 clients, and championed innovation in digital experience technologies.

Dr. Roger E. Ari Kahn – President & Chief Executive Officer



Dr. Roger E. Kahn joined Bridgeline as Chief Operating Officer in August 2015 and was named President & CEO in May 2016. Holding a Ph.D. in Computer Science & Artificial Intelligence from the University of Chicago, he co-founded FatWire—pioneering web content management and digital engagement—scaling it to a \$40 million business across 13 countries before its 2011 acquisition by Oracle. At Bridgeline, Dr. Kahn has driven the shift to AI-powered search and unified digital experience solutions, expanded the company’s enterprise customer base, and instilled a culture of customer-focused innovation and measurable ROI

CHAPTER 3

SCOPE OF WORK

The scope of work for my internship at Bridgeline focused on extending the eCommerce 360 dashboard and HawkSearch integration by delivering both frontend and backend enhancements. Key areas included React-based widget development, AI-driven search pipeline configuration, REST API integration, and systematic bug fixes to improve user experience, search relevancy, and analytics.

1. Frontend Validation and Error Handling

- **Objective:**
To create interactive dashboard components that visualize search performance and traffic metrics, and to improve overall usability.
- **Scope:**
 - Built new React widgets for zero-result alerts, top-search terms, and conversion funnels.
 - Added client-side validation and error states for missing data or API timeouts.
- **Methodology:**
Implemented conditional checks within React components to restrict unsupported file types and displayed real-time error messages for invalid user inputs.
- **Results:**
Improved frontend reliability, reduced incorrect file uploads, and provided a smoother user experience through responsive error notifications.

2. Backend Logic Implementation

- **Objective:**
To customize HawkSearch pipelines for concept and image search, and expose those endpoints to the frontend.
- **Scope:**
 - Defined custom synonym rules and relevance weights in the HawkSearch management console.
 - Documented API contracts and sample payloads for the frontend team.
- **Methodology:**
 - Collaborated with the search-ops team to translate business rules into pipeline configurations.

- Used Postman to prototype and validate each new endpoint, then coded the Express routes and controllers.
- **Results:**
Achieved efficient backend functionality with reduced error rates and seamless communication between client and server.

3. Testing and Bug Fixes

- **Objective:**
To ensure stability across the dashboard and search features through comprehensive testing and bug resolution.
- **Scope:**
 - Identified cross-browser layout issues, race conditions in API calls, and edge-case errors in search filters.
 - Wrote unit tests for React components and integration tests for backend endpoints.
- **Methodology:**
Conducted testing using Postman for backend endpoints, used browser dev tools for frontend debugging, and collaborated with team members to isolate and fix bugs.
- **Results:**
Enhanced application stability, reduced form handling errors, and ensured smoother development cycles through systematic debugging.

CHAPTER 4

LEARNING OBJECTIVES

1. Strengthen Full-Stack Development Skills

Leverage a real-world codebase to deepen proficiency with React.js (Hooks, Context API, component lifecycle) and Node.js/Express backend services. I practiced end-to-end feature delivery—building responsive dashboard widgets, implementing RESTful endpoints, and wiring CRUD operations against MongoDB Atlas—solidifying your understanding of how UI components, API layers, and databases collaborate in a modern tech stack.

2. Master AI-Powered Search Integration

Gain hands-on experience configuring and tuning HawkSearch pipelines for concept- and image-based queries. I learned to translate business requirements into relevance rules, synonyms, and boosting strategies via the HawkSearch management console, then exposed those capabilities through new API routes. This objective deepened your grasp of how machine-learning models drive personalized search experiences at enterprise scale.

3. Enhance Data Analytics & Visualization

Develop the ability to turn raw event data into actionable insights by building interactive React widgets for the eCommerce 360 dashboard. You designed components to display zero-result trends, top search terms, and conversion funnels, and implemented aggregation pipelines in MongoDB to pre-compute metrics. This work sharpened your skills in data modeling, performance tuning, and user-centric visualization of key performance indicators.

4. Develop Robust Error Handling and File Management Skills

Ensuring secure and error-free user experiences was another important learning objective. I implemented form validation mechanisms on the frontend to control user inputs and prevent improper data submission. I also worked on restricting file uploads to specific formats (such as images and JSON), thereby avoiding potential vulnerabilities. This involved using React Toastify for real-time error alerts and FastAPI backend logic for server-side validations. These tasks helped me understand how critical proper input validation and file handling are to prevent issues like broken APIs, insecure uploads, or inconsistent data especially in cybersecurity tools dealing with sensitive reports.

5. Gain Exposure to Real-World Project Management

Lastly, the internship provided me with an opportunity to observe and participate in real-world project management workflows. I was included in planning discussions, feature reviews, and internal testing phases. I worked closely with mentors and team members to fix bugs, update UI components, and refine user interactions based on feedback. I maintained proper folder structures, consistent naming conventions, and development notes to ensure code clarity and maintainability. Through this experience, I gained a deep appreciation for agile development practices and learned how even minor UI/UX changes can significantly improve product usability in professional environments.

CHAPTER 5

CHALLENGES AND SOULTION

1. Understanding a Large Production-Level Codebase

Challenges:

The HawkSearch React SDK architecture was complex, with deeply nested component hierarchies and little initial documentation, making it difficult to onboard and implement new features quickly.

Solutions:

Spent dedicated time exploring the codebase, tracing component relationships, leveraging browser and IDE debugging tools, and studying existing logic flows to build a clear mental model of the system.

2. Identifying UI and API Behavior Issues

Challenges:

Certain API responses did not surface correctly in the UI, leading to mismatches between backend data and what users saw on screen.

Solutions:

Used browser dev tools to capture and inspect network calls, compared expected versus actual payloads, and debugged React state management to ensure data was parsed and rendered accurately.

3. Issues Not Saving Automatically

Challenges:

Users often forgot to manually save their inputs when documenting findings or editing issue fields, which led to frequent data loss. This was particularly problematic during multitasking or lengthy analysis sessions, where unsaved work could be lost accidentally.

Solutions:

An auto-save feature was integrated into the portal using JavaScript triggers and backend API calls to save form data periodically or when fields were modified. This greatly improved user experience by ensuring that progress was retained without requiring constant manual intervention, boosting productivity and reducing frustration.

5. Bug Localization and Fixing

Challenges:

Asynchronous data flows and state updates triggered elusive bugs and unintended re-renders, complicating debugging and prolonging development cycles.

Solutions:

Introduced granular logging around asynchronous operations, inserted breakpoints in critical lifecycle methods, and monitored state transitions to pinpoint mismatches and apply precise fixes.

6. Adapting Features to Varied Client Requirements

Challenges:

Client proof-of-concepts demanded different search behaviors and UI configurations, requiring frequent, ad-hoc code changes that risked introducing regressions.

Solutions:

Refactored the SDK integration into modular, configurable components and created parameter-driven prototypes, enabling rapid adjustments without altering core logic or impacting existing features.

7. Difficulty Learning React

Challenges:

As someone relatively new to the tech stack, initial learning curves with React (frontend). Understanding component lifecycles, hooks, state management in React, and asynchronous routing in FastAPI was challenging without prior experience.

Solutions:

To overcome this, I adopted a hands-on learning approach by building mini-projects using both frameworks in isolation. This helped me gradually understand their inner workings, structure, and best practices. Team mentorship and code reviews also played a key role in accelerating my learning curve and improving code quality over time.

8. Complex and Confusing Project Structure

Challenges:

The original folder and file organization in the project lacked clarity, with overlapping naming conventions and minimal internal documentation. This made onboarding, navigation, and debugging tedious, especially when working across multiple modules.

Solutions:

I reorganized the project files based on functional domains and added developer notes for easy reference. Naming conventions were standardized to ensure clarity and modularity. These changes significantly improved code maintainability and made the codebase more approachable for new contributors and reviewers.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

Operating System: Windows 10/11

Development Tools:

- Visual Studio Code
- Git
- Postman

Programming Languages and Frameworks:

Frontend:

- Language: JavaScript
- Framework: React.js and Angular (used for UI development and component-based structure)
- Libraries Used:
 - Axios – for HTTP requests
 - Tailwind CSS (if used) – for quick UI styling and responsive design

Backend:

- Language: C#
- Framework: dotNet core

Database: MySQL

Runtime and Package Managers:

- Node.js and npm – For managing JavaScript packages and running React dev.

Hardware requirements:

- Processor: Intel i5 or AMD Ryzen 5 and above
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: At least 256 GB SSD
- Internet: Stable high-speed connection

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

During my internship at Bridgeline, I had the opportunity to make impactful contributions to the e360 Site Performance Report and the AI Agentic Configuration Platform. By delivering both frontend and backend enhancements, tuning HawkSearch pipelines, integrating analytics, and streamlining development workflows, I helped elevate platform usability, performance, and configurability.

1. Frontend Enhancements

One of my primary focuses was improving the user interface and experience of the e360 dashboard and adding concept and image features into Hawksearch reactSDK.

- **UI/UX Improvement:** I redesigned certain components to ensure consistency in layout, color schemes, and element placement. The goal was to reduce visual clutter and simplify the user experience, especially during the issue submission process.
- **Image Format Validation:** To improve file input reliability, I added a validation mechanism to accept only specific image formats (JPEG, JPG, PNG). This reduced the risk of incompatible file types being uploaded, ensuring smooth image rendering and improving backend processing.
- **Real-time Feedback with Toast Messages:** To give users immediate visual cues on their actions (e.g., form submission, errors, successful uploads), I integrated toast messages using a notification library. This significantly enhanced interaction feedback, helping users stay informed about the status of their actions without navigating away from the page.

2. Backend Improvements

On the server side, I strengthened API endpoints and data-handling logic to support new frontend features and ensure robust performance.

- **API Development:** Added and documented five new REST routes in the Node.js/Express backend for retrieving pre-aggregated e360 metrics, triggering HawkSearch concept/image queries, and submitting AI configuration changes.

- **Caching & Rate-Limit Handling:** Implemented in-memory caching with a sliding-window eviction policy to reduce redundant HawkSearch calls on high-traffic endpoints, and added exponential-backoff retry logic for rate-limit resilience.
- **Auto-Archive of Analytics:** Introduced a nightly batch job using cron and RabbitMQ to move records older than 30 days from the primary collection into an archive database, improving query performance on current data.

3. Development, Testing, and Deployment Workflow

This phase of my work revolved around ensuring that the implemented features were stable, efficient, and well-integrated across the system.

- **Full-Stack Feature Development:** I built and tested key features combining **React (frontend)**, **FastAPI (backend)**. This provided a hands-on opportunity to apply my learning in a real-world development setting.
- **Cross-Component Issue Validation:** I worked on verifying data flow and issue creation logic across frontend forms, backend APIs, and the database, ensuring smooth interaction and error-free submissions.
- **Code Optimization:** While reviewing the existing codebase, I identified unused or redundant fields/components in the UI and removed them to improve performance and reduce cognitive load for users.
- **Debugging & Git Workflow:** I regularly debugged features in the DEV environment, using browser tools, logs, and API traces. For collaboration and version control, I followed a Git-based workflow. I used **GitLab** to manage feature branches, conducted peer reviews, and staged changes before merging. Once validated, the final changes were pushed to **GitHub** for deployment.

Technology stack

Frontend Technologies

- **React.js:** A popular JavaScript library used to build dynamic and responsive user interfaces. It enabled the development of reusable components and efficient rendering, improving the overall user experience.
- **JavaScript, HTML5, CSS3:** These standard web technologies were used to structure the application layout, implement logic, and apply styling to make the interface user-friendly and visually appealing.

Backend Technologies

- **dotNet Core(C#):** A high-performance web framework used to build RESTful APIs. It enabled fast development with automatic data validation, asynchronous programming, and clear routing mechanisms to handle HTTP requests efficiently.

API Testing

- **Postman:** Used to test backend API endpoints during development. It allowed sending custom HTTP requests, validating responses, and identifying bugs before integrating with the frontend.

Version Control

- **Git (GitHub):** Version control was managed using Git. GitHub was used during the development phase for branching, staging, and merge requests,

HTTP Requests

- **Axios:** A promise-based HTTP client used in React for sending asynchronous requests to the FastAPI backend. It enabled easy integration with API endpoints and efficient data handling between client and server.

Development Tools

- **VS Code:** The primary code editor used for writing and organizing code. It offered useful extensions for linting, formatting, and Git integration.
- **Terminal & Git Bash:** Used for command-line operations like running servers, pushing code, and managing version control through Git commands.

Development Workflow And Process

Through these contributions, I not only enhanced the functionality and reliability of Bridgeline's e360 and HawkSearch platforms but also honed my full-stack, AI-integration, and DevOps skills in a fast-paced MarTech environment.:

- **Problem Identification and Solution Design**

Each task began with identifying a problem or understanding a new feature request. These were typically raised during team discussions or by proactively noticing gaps or limitations in the current implementation. Examples included user interface issues, missing validations, bugs in API behavior, or feature enhancement requirements. After identifying the core problem, I designed a solution considering user needs, technical feasibility, and simplicity. I documented the intended logic, outlined the affected components, and clarified inputs and outputs to avoid rework later in the process.

- **Code Implementation – Frontend and Backend**

After designing the solution, I began the development phase, which involved both frontend and backend components:

- **Frontend (React.js):** I created or updated UI components using React. This included modifying layouts, adding new form fields, handling state, and ensuring a consistent and responsive design. I used libraries like Formik and Yup for form handling and validation to provide a smoother user experience.
- **Backend (FastAPI):** On the backend, I developed RESTful API endpoints using FastAPI. These endpoints handled operations such as data fetching, creation, updates, and deletions. I implemented input validations and error handling, and ensured secure interactions with the MongoDB database using asynchronous calls (via Motor). I kept the code modular, with clearly defined routes and database service layers.

- **API Testing Using Postman**

Before integrating with the frontend, I tested each backend API using Postman. I verified that all endpoints responded correctly to valid inputs and handled invalid or missing data with appropriate error messages. This step ensured that the logic on the backend was functioning correctly in isolation, which made the later frontend integration smoother and helped avoid runtime bugs.

- **Testing in Localhost and DEV Environment**

After successful backend testing, I ran the full stack locally on my development machine to simulate actual user interaction. This allowed me to verify the integration between frontend and backend, validate data flow across components, and check for UI or usability issues. Once everything worked as expected locally, I deployed the changes to the DEV environment used by the team. This environment mirrored the production setup, and testing here helped identify environment-specific issues such as CORS policies, routing behavior, or performance inconsistencies.

- **Review and Approval Process**

When development and testing were complete, I pushed my feature or bug-fix branch to the remote GitLab repository. I raised a Merge Request (MR) linked to the specific issue or feature. The MR was reviewed by senior developers or team leads who checked the logic, code readability, naming conventions, and whether best practices were followed. They sometimes provided suggestions or asked for changes, which I implemented before the code was finally approved. This review process ensured code quality and team-wide standards were maintained.

- **Staging and Merging Using GitHub**

After receiving approval:

- I staged all validated changes and ensured the branch was up to date with the latest development branch to avoid conflicts.
- I submitted a final Merge Request with comments explaining the changes and references to issue numbers.
- The QA or tech lead did a final validation of the merged code in the staging environment.
- Once verified, the MR was merged into the main development branch.

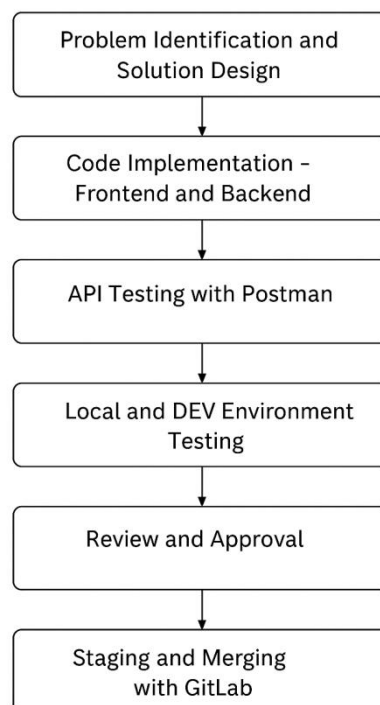


Fig 6.1: Development and Testing

CHAPTER 7

REFLECTION AND ANALYSIS

The internship at Bridgeline offered a comprehensive immersion into enterprise-scale MarTech engineering—combining full-stack development, AI-driven search configuration, data analytics, and DevOps practices. It was not merely an opportunity to learn new technologies, but also to understand how a unified digital experience platform translates technical features into measurable business outcomes. From building interactive dashboard widgets and tuning HawkSearch pipelines to orchestrating CI/CD workflows, this experience fostered both technical mastery and professional maturity.

During this period, I built a solid foundation in Node.js/Express and MongoDB Atlas, learning how to structure RESTful services, implement caching strategies, and design aggregation pipelines for real-time analytics. On the frontend, I honed my React.js skills—using Hooks, Context API, and Recharts—to create dynamic, responsive components for zero-result alerts, top-search term charts, and conversion funnels.

To apply my learning in isolation, I developed proof-of-concept demos—a standalone concept-search widget and an interactive analytics prototype—which helped me validate search relevancy rules and event ingestion flows before integrating them into the e360 dashboard and AI Agentic Configuration Platform. These mini-projects reinforced best practices in component design, data validation, and performance tuning.

I gained valuable insight into the MarTech domain by participating in product demos and proof-of-concept sessions, seeing firsthand how Bridgeline’s “Grow Your Online Revenue” framework links traffic acquisition, conversion optimization, and average order value uplift to features like real-time indexing, machine-learning relevance tuning, and merchandising controls. Understanding this end-to-end revenue driver model deepened my appreciation for technology’s role in business strategy.

Collaborating with cross-functional teams sharpened my professional skills. I engaged in agile ceremonies, wrote detailed merge requests in GitLab, and configured Jenkins

pipelines for linting, testing, and blue/green deployments. Regular code reviews taught me the importance of clear documentation, modular design, and iterative improvement to maintain high-quality releases.

Debugging complex, asynchronous flows—spanning frontend state, backend routes, and message queues—honed my investigative approach. By leveraging contract-first API definitions, Postman mocking, and targeted logging, I learned to isolate and resolve elusive issues efficiently, ensuring robust end-to-end functionality.

In conclusion, this internship has been transformational—equipping me with robust full-stack, AI integration, and DevOps capabilities, alongside deep MarTech domain knowledge. Bridgeline confident in my ability to architect scalable, data-driven platforms that deliver tangible business impact and ready to contribute effectively in any enterprise software environment.

CHAPTER 8

RECOMMENDATIONS

1. Introduce Well-Defined Internship Structures

Bridgeline can enhance future internship programs by adopting a clear framework that outlines roles, objectives, timelines, and deliverables from day one. Providing interns with a structured roadmap—combining hands-on projects, scheduled check-ins, and milestone reviews—will help them stay focused, monitor their progress, and achieve meaningful outcomes.

2. Foster Academic Collaborations

Establish partnerships with universities and technical institutes to bridge theoretical learning with real-world MarTech challenges. Joint initiatives—such as guest lectures by Bridgeline experts, hackathons focused on AI-driven search, or collaborative research projects on digital experience optimization—can enrich the internship experience and position Bridgeline as a talent incubator.

3. Establish Strong Mentorship Channels

Assigning each intern a mentor who offers technical guidance and regular feedback can make the learning curve smoother. Structured mentor–intern check-ins, feedback reviews, and support can help interns better understand workflows and boost their confidence in applying tools and solving problems independently.

4. Encourage Active Participation and Creative Problem Solving

Creating opportunities for interns to contribute ideas, participate in meetings, and take ownership of smaller projects encourages innovation. This involvement not only enhances their learning but also brings fresh ideas to ongoing work, fostering a two-way benefit between the intern and the organization.

5. Create Networking and Exposure Opportunities

Facilitating cross-functional interactions can help interns explore various career paths. Organizing Q&A sessions, informal chats with professionals, or internal demo days allows interns to learn beyond their immediate teams, increasing awareness of the broader tech and cybersecurity landscape.

CHAPTER 9

CONCLUSION

My internship journey at Bridgeline has been an incredibly enriching and transformative experience. Over the course of this internship, I not only gained technical knowledge but also developed a deeper understanding of how real-world MarTech projects operate in an enterprise environment. From learning Express and MongoDB Atlas on the backend to applying core React.js concepts on the frontend, I had the opportunity to work across the full technology stack. This end-to-end exposure—from configuring HawkSearch pipelines and building e360 dashboard widgets to tuning AI agentic configurations—helped me see how disparate technologies integrate to deliver robust, revenue-driving digital experiences.

One of the highlights of my time at Bridgeline was building focused mini-projects that preceded larger feature work. I created a standalone concept-search prototype and an interactive analytics demo to validate relevance tuning and event-ingestion flows before integrating them into the main e360 Site Performance Report and AI Configuration Platform. These exercises reinforced my skills in component design, API communication, and performance optimization, and laid the groundwork for the full-stack enhancements I later contributed.

Beyond the technical work, this internship helped me grow professionally. I learned the importance of agile collaboration—breaking requirements into user stories, estimating effort, and adapting to shifting priorities. Debugging complex asynchronous flows across React state, Express middleware, and message queues sharpened my investigative approach. Managing my time across multiple sprints, responding to bug fixes, and maintaining a clean project structure taught me how to balance speed with code quality in a production setting.

Moreover, the mentorship and guidance I received made a significant difference. Regular one-on-one check-ins, code reviews, and demo presentations provided invaluable feedback that accelerated my learning curve. Discussing architectural trade-offs with senior engineers and observing best practices in version control and CI/CD strengthened my confidence and equipped me to tackle increasingly complex tasks independently.

In conclusion, my time at Bridgeline has been a pivotal stepping stone in my career. It has bolstered my full-stack development and MarTech platform skills, deepened my understanding of AI-driven search and analytics, and instilled professional habits that will serve me well in any software engineering role.

CHAPTER 10

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